

GVPT Data Science Camp

Data Presentation

Ryan Bookstein

August 27, 2026

University of Maryland

Learning Objectives

- ▶ Learn how to combine code, results, and figures into one document

Learning Objectives

- ▶ Learn how to combine code, results, and figures into one document
- ▶ Introduction to markdown, Quarto, Overleaf, and $\text{\LaTeX}$

Quarto and Markdown

The screenshot displays the Quarto application interface, which is used for creating documents that weave together content and executable code. The interface is divided into several panes:

- Source Pane (Left):** Contains the source code for the document. It includes a YAML header with metadata (title, format, editor) and R code chunks for loading packages and plotting.


```
---
title: "Hello, Quarto"
format: html
editor: visual
---

{r}
#| label: load-packages
#| include: false

library(tidyverse)
library(palmerpenguins)

{r}
#| label: plot-penguins
```
- Render Pane (Top Right):** Shows the rendered output of the document. It includes the title "Hello, Quarto", a section "Meet Quarto" with a link to the Quarto website, and a section "Meet the penguins" with a description of the penguins data and a plot.

Meet the penguins

The `penguins` data from the `palmerpenguins` package contains size measurements for 344 penguins from three species observed on three islands in the Palmer Archipelago, Antarctica.

The plot below shows the relationship between flipper and bill lengths of these penguins.

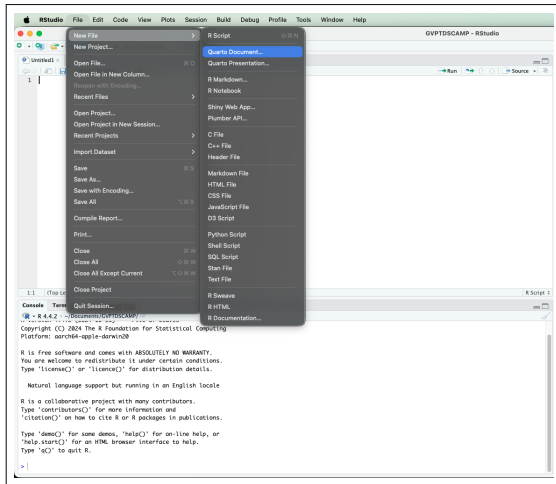
Flipper and bill length
Dimensions for penguins at Palmer Station LTER

The plot is a scatter plot showing the relationship between flipper length (mm) on the x-axis and bill length (mm) on the y-axis. The data points are colored by penguin species: Adelia (orange), Chinstrap (purple), and Gentoo (green). The plot shows a positive correlation between flipper length and bill length for all three species.
- Environment Pane (Bottom Right):** Shows the environment of the R session, including the loaded packages and their versions.
- Console Pane (Bottom Left):** Shows the output of the R code chunks.

Source: Quarto

- ▶ A simple and easy-to-use markup language to format your documents

Create a Quarto Document



Create a Quarto Document

New Quarto Document

 Document
 Presentation
 Interactive

Title:

Author:

☒ **HTML**
Recommended format for authoring (you can switch to PDF or Word output anytime)

☐ **PDF**
PDF output requires a LaTeX installation (e.g. <https://yihui.org/tinytex/>)

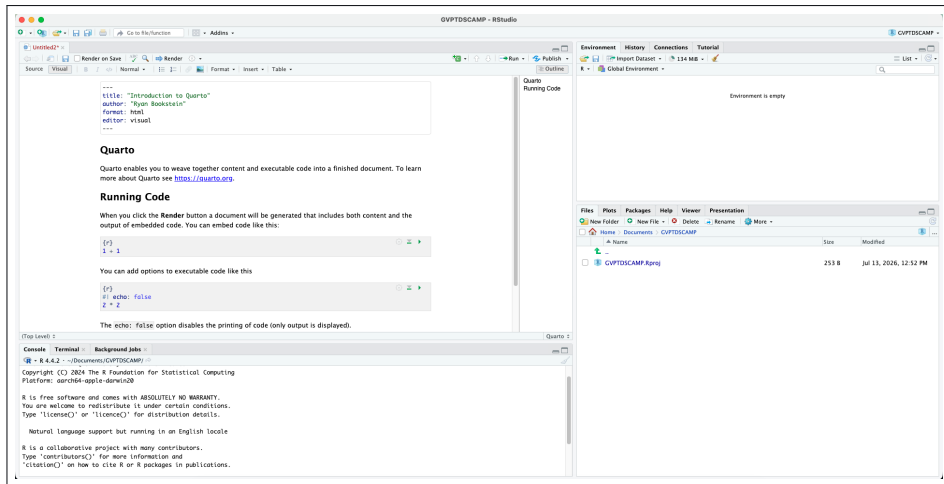
☐ **Word**
Previewing Word documents requires an installation of MS Word (or Libre/Open Office on Linux)

Engine:

Editor: ☒ Use visual markdown editor 

 [Learn more about Quarto](#)

Create a Quarto Document



- ▶ Every Quarto document opens with a header between two rows of `---` that sets the title, author, and output format

```
---  
title: "Introduction to Quarto"  
author: "Your Name"  
format: html  
---
```

- ▶ Every Quarto document opens with a header between two rows of `---` that sets the title, author, and output format

```
---  
title: "Introduction to Quarto"  
author: "Your Name"  
format: html  
---
```

- ▶ Change `format: html` to `format: pdf` for a PDF

- ▶ Rendering to PDF needs a $\text{\LaTeX}$ installation. Run this once, in the RStudio console:

```
install.packages("tinytex")  
tinytex::install_tinytex()
```

- ▶ Rendering to PDF needs a $\text{\LaTeX}$ installation. Run this once, in the RStudio console:

```
install.packages("tinytex")  
tinytex::install_tinytex()
```

- ▶ Restart RStudio afterwards. HTML output needs nothing extra

Rendering Your Document

The screenshot displays the RStudio interface with a Quarto document open. The editor on the left shows the source code of the document, which includes a title, author, and several code blocks. The right pane shows the rendered HTML output, which includes the title, author, and sections for 'Quarto' and 'Running Code'. The bottom console shows the R environment output, including the R version and the location of the document.

```
1 ---
2 title: "Introduction to Quarto"
3 author: "Ryan Bookstein"
4 format: html
5 editor: visual
6 ---
7
8 # Quarto
9
10 Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see
11 <https://quarto.org>.
12
13 # Running Code
14
15 When you click the Render button a document will be generated that includes both content and the output of embedded code. You can
16 embed code like this:
17
18 ```{r}
19 1 + 1
20 ```
21
22 You can add options to executable code like this
23
24 ```{r}
25 #| echo: false
26 2 + 2
27 ```
28
29 The 'echo: false' option disables the printing of code (only output is displayed).
```

Introduction to Quarto

AUTHOR
Ryan Bookstein

Quarto

Quarto enables you to weave together content and executable code into a finished document. To learn more about Quarto see <https://quarto.org>.

Running Code

When you click the **Render** button a document will be generated that includes both content and the output of embedded code. You can embed code like this:

```
1 + 1
```

[1] 2

You can add options to executable code like this

```
[1] 4
```

The `echo: false` option disables the printing of code (only output is displayed).

R 4.4.2 (2024-04-24) (64-bit)
type 'license()' or 'licence()' for distribution details.

Natural language support but running in an English locale

R is a collaborative project with many contributors.
Type 'contributors()' for more information and
'citation()' for how to cite R or R packages in publications.

Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> |

- ▶ Code chunks start with ````{r}` and end with `````

```
```{r}
#| echo: true

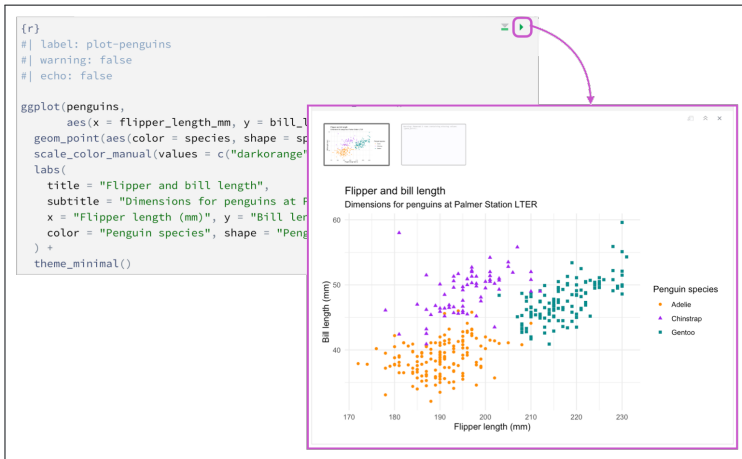
library(tidyverse)
```
```

- ▶ You can control how code chunks render and are displayed

```
```{r}  
#| echo: true
#| warning: false

library(tidyverse)
```
```


Code Chunks in Quarto



Source: Quarto

Overleaf and $\text{\LaTeX}$

- ▶ Overleaf is an online $\text{\LaTeX}$ editor used to write, edit, and publish scientific documents

- ▶ Overleaf is an online $\text{\LaTeX}$ editor used to write, edit, and publish scientific documents
- ▶ $\text{\LaTeX}$ is a high-quality document preparation system and markup language

Creating an Overleaf Account

- ▶ Open your browser and navigate to overleaf.com/edu/umd

Creating an Overleaf Account

- ▶ Open your browser and navigate to overleaf.com/edu/umd
- ▶ Click Log In or Register

Creating an Overleaf Account

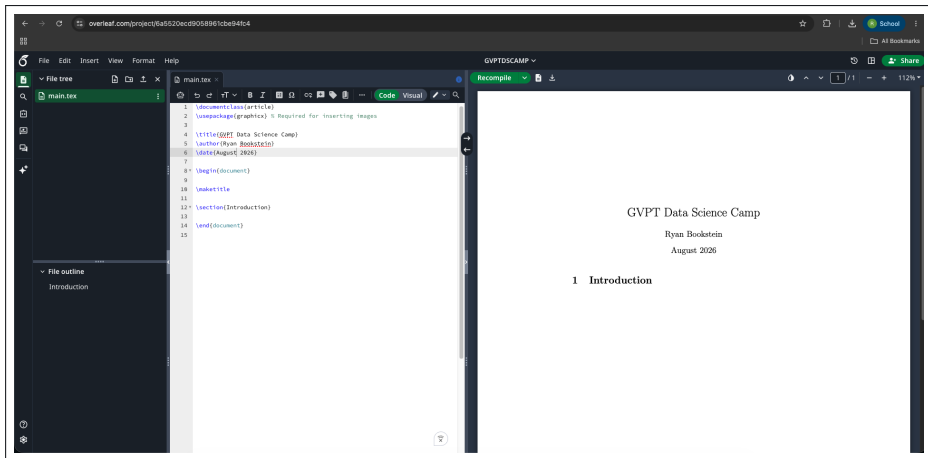
- ▶ Open your browser and navigate to overleaf.com/edu/umd
- ▶ Click Log In or Register
- ▶ Select the option to login through your institution (SSO) and use your UMD Directory ID and password to link your account

- ▶ Navigate to your project dashboard

- ▶ Navigate to your project dashboard
- ▶ Select New project, Blank project

- ▶ Navigate to your project dashboard
- ▶ Select New project, Blank project
- ▶ Name it

- ▶ Navigate to your project dashboard
- ▶ Select New project, Blank project
- ▶ Name it
- ▶ Select Create



Setting Up Your Document

- ▶ Every Overleaf document requires a “preamble” at the top to define the page layout

```
\documentclass{article}

\title{GVPT Data Science Camp}
\author{Ryan Bookstein}
\date{August 2026}

\begin{document}
\maketitle

\section{Introduction}
Hello World! This is my first Overleaf document.

\end{document}
```

- ▶ Unlike other word processors, you do not highlight text to format it. Wrap the text in commands to achieve desired formatting

```
\textbf{GVPT Data Science Camp.}
```

```
\textit{GVPT Data Science Camp.}
```

- ▶ Keep your document organized by using built-in sectioning commands

```
\section{Introduction}  
This is the introduction.  
  
\subsection{Background}  
Here is some background information.
```

- ▶ Combined code, results, and figures in a single Quarto document

- ▶ Combined code, results, and figures in a single Quarto document
- ▶ Rendered that document to HTML and PDF

- ▶ Combined code, results, and figures in a single Quarto document
- ▶ Rendered that document to HTML and PDF
- ▶ Set up an Overleaf account and wrote your first $\text{\LaTeX}$ document

- ▶ Build your presentation for this afternoon in either Quarto or Overleaf — whichever you found more comfortable

- ▶ Build your presentation for this afternoon in either Quarto or Overleaf — whichever you found more comfortable
- ▶ Bring the rendered file, not just the source

Questions?